

TITAN 11

MASTER SINGLE SOURCE OF TRUTH

v1.0 (HARMONIZED)

STATUS: LOCKED (Architectural)

SSOT_VERSION:	1.0
SYSTEM_NAME:	TITAN 11 Enterprise Decision Intelligence Platform
CONFIDENCE:	0.97
LAST_UPDATED:	2026-04-26
DOCUMENT_TYPE:	Canonical SSOT for EIB / EBRD / Internal Governance / Audit Trail

This document constitutes the authoritative, immutable Single Source of Truth (SSOT) for the TITAN 11 system architecture, governance model, decision logic, and operational protocols. It is designed for regulatory submission, board review, and enterprise audit.

1. SYSTEM IDENTITY

Field	Value
System Type	Decision Intelligence Platform
Domain	Financial + Document Intelligence
Deployment Model	Local + Docker + Enterprise API
Target Users	CFO / Lenders / Analysts / Admin
Audit Level	Audit-grade (deterministic core)
SSOT Role	Authoritative Truth Layer

2. CORE PIPELINE (CANONICAL FLOW)

RAW → PARSING → EXTRACTION → INGESTION → SEMANTICS → ENTITY → RELATION → IMPORTANCE → RULES → AUDIT → LOGIC → SSOT → SNAPSHOT → SUBMISSION → RBAC → EXPORT → API → UI → DEPLOYMENT

3. LAYER ARCHITECTURE (LOCKED)

LEVEL 0 — INGESTION

document scanners • job queue + heartbeat • incremental ingestion • file hashing (content-based)

LEVEL 1 — PARSING

PDF / DOCX / XLSX extraction • OCR fallback • layout-aware parsing

LEVEL 2 — SIGNAL EXTRACTION

numeric normalization (locale-aware) • financial signals • structural signals

LEVEL 3 — SEMANTICS

role classification (data / rule / evidence) • entity extraction (who / what / when / value)

LEVEL 4 — RELATION

entity linking • cross-document consistency

LEVEL 5 — PRIORITY

importance scoring • noise vs signal separation

LEVEL 6 — RULE ENGINE

implicit logic inference • financial rules • constraint validation

LEVEL 7 — SSOT (CRITICAL)

conflict resolution (weighted authority) • canonical record creation • confidence scoring

LEVEL 8 — AUDIT

anomaly detection • consistency validation • lineage tracking

LEVEL 9 — DECISION

readiness scoring • risk scoring • simulation engine

LEVEL 10 — GOVERNANCE

snapshot (immutable) • lender submission mode • RBAC approval

LEVEL 11 — DELIVERY

document pack export • Excel / PDF / Word outputs

LEVEL 12 — PLATFORM

API layer (FastAPI) • UI layer (Streamlit → future React) • Docker deployment

4. SSOT LOGIC (CRITICAL CORE)

Conflict Resolution Model

Final_Value = argmax(Σ authority_weight × signal_consistency)

Authority Hierarchy (LOCKED)

Rank	Source
1.0	Signed / Final Documents
0.9	Official Financial Statements
0.8	Internal Validated Reports
0.6	Draft Documents
0.4	Derived Signals
0.2	Heuristic / AI

Confidence Rules

Condition	Status
≥ 0.90	LOCKED
0.70 – 0.89	REVIEW
< 0.70	CONFLICT

5. GOVERNANCE MODEL

Submission Lifecycle

LOCKED_FOR_REVIEW → APPROVAL_REQUESTED → APPROVED / REJECTED → EXPORT_ALLOWED

Mandatory Conditions for Lender Export

1. SSOT snapshot exists
2. Submission record exists
3. Approval event exists
4. Role = CFO or ADMIN

6. RBAC MODEL

Role	Permissions
ADMIN	full control
CFO	approve + export
ANALYST	read + write
VIEWER	read
LENDER	read_lender_pack

7. DECISION ENGINE MODEL

Final Score Formula

$$\text{FINAL_SCORE} = 0.5 \times \text{Readiness} + 0.3 \times (1 - \text{Risk}) + 0.2 \times (1 - \text{Anomaly})$$

Decision Thresholds

Score	Decision
≥ 0.85	READY
0.65 – 0.84	CONDITIONAL
0.45 – 0.64	NOT READY
< 0.45	BLOCK

8. ANOMALY FRAMEWORK

Detected Types

- SSOT_CONFLICT
- NEGATIVE_REVENUE
- CONFIDENCE_MISMATCH
- RISK_URGENCY_MISMATCH

Severity Mapping

Severity	Score
LOW	< 0.4
MEDIUM	0.4 – 0.7
HIGH	> 0.7

9. INFRASTRUCTURE

Runtime

Python 3.11 • SQLite (WAL mode) • FastAPI • Streamlit • Docker

Key Features

- idempotent processing
- crash-safe queue
- heartbeat recovery
- snapshot immutability

10. ENTERPRISE PLATFORM

- Auth API (JWT)
- Workflow API
- Decision Engine
- Dashboard UI
- Deployment Pack

11. SYSTEM CLASSIFICATION

TYPE:	Enterprise Financial Intelligence System
CATEGORY:	Decision Support + Audit Platform
MATURITY:	Production-Ready (Single-node)
UPGRADE PATH:	Distributed + Cloud-native

12. KNOWN GAPS (NON-BLOCKING)

Area	Gap
AI	anomaly ML model
UI	full React frontend
Auth	password hashing
Infra	distributed DB
Monitoring	Prometheus

13. FINAL SSOT STATEMENT

TITAN 11 is a deterministic, audit-grade decision intelligence system that transforms unstructured documents into validated financial truth, governed through SSOT, and controlled via enterprise workflow before lender submission.



Document Control

This SSOT is immutable. All changes require formal Change Control Board approval.
Prepared for: EIB / EBRD Submission | Internal Governance | Audit Trail | System Control Tower

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